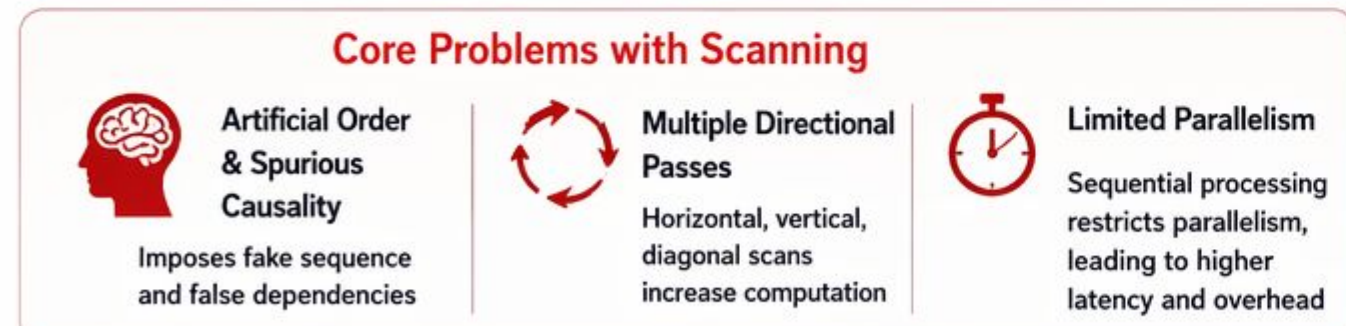
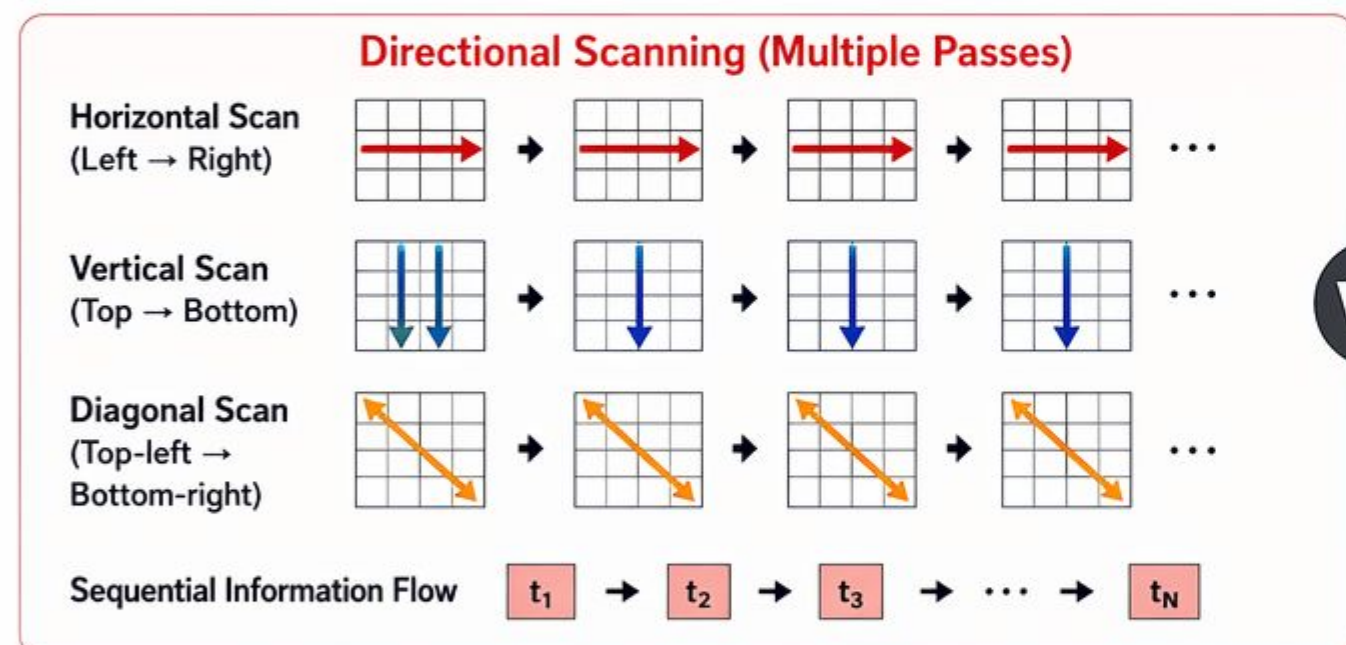
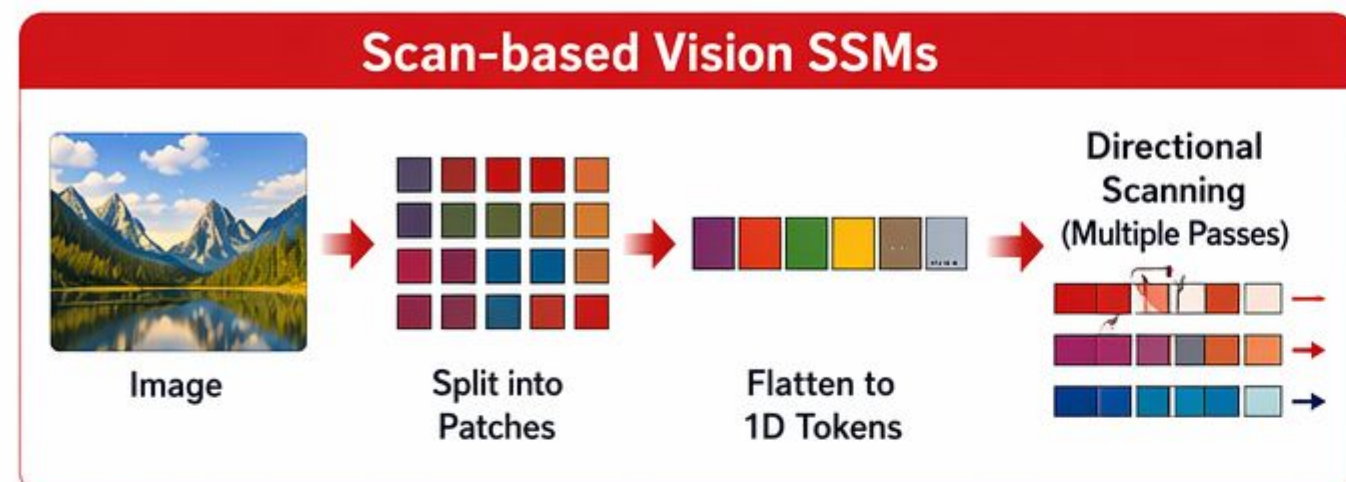


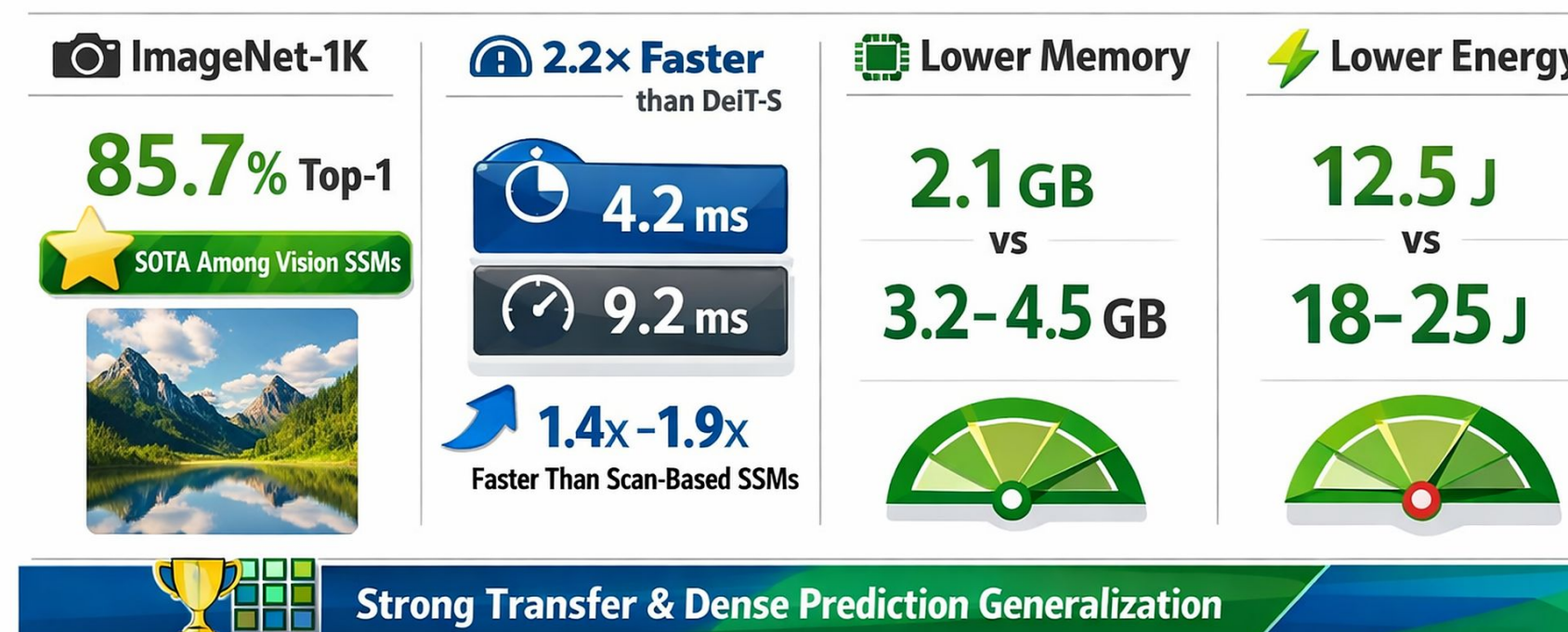
Scan-based Vision SSMs vs HAMSA

Same Patching, Different Processing: Sequential Scans vs Global Spectral Interaction



All results measured on ImageNet-1K (batch size = 128). Higher is better for Top-1 Accuracy and Throughput. Lower is better for Latency, Memory, and Energy.

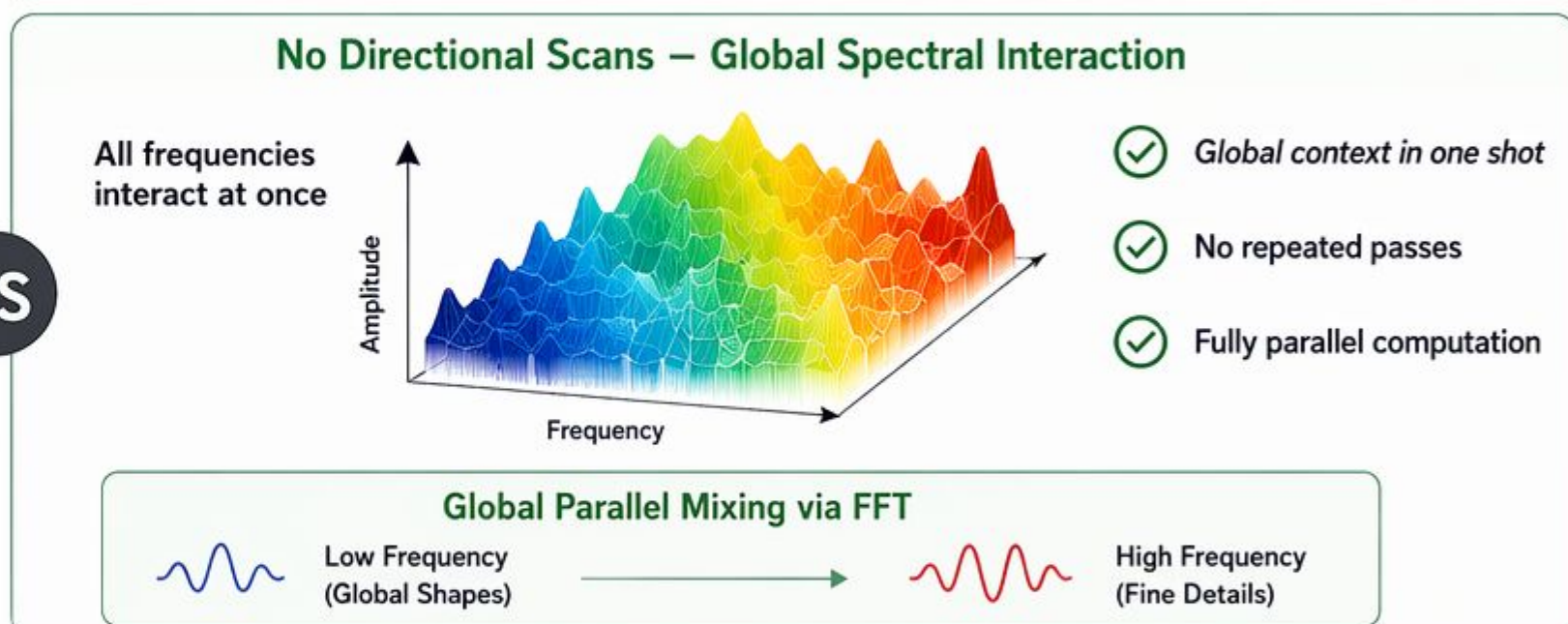
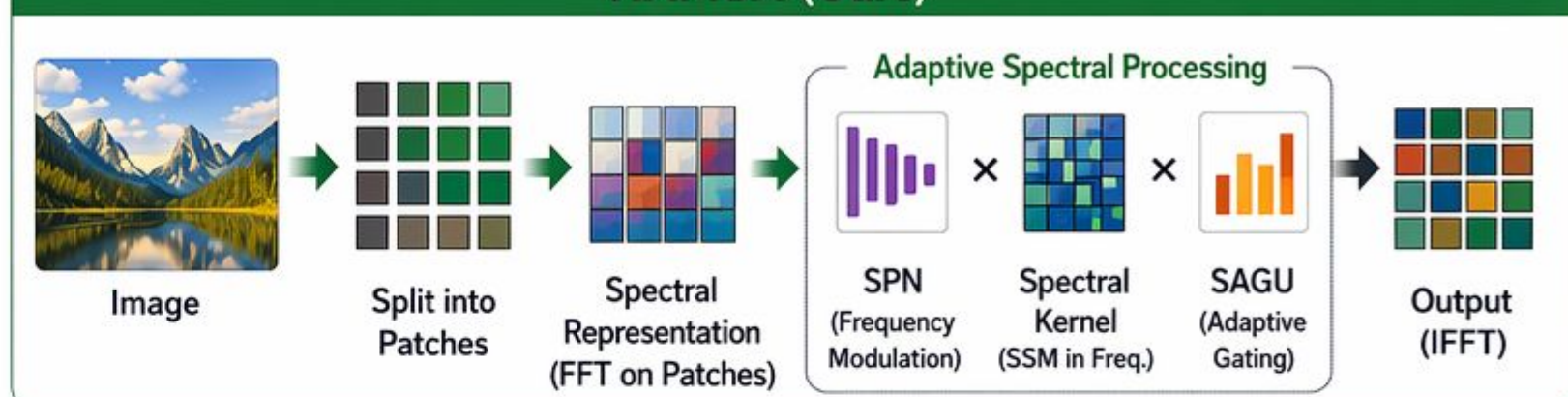
State of the Art + Efficiency



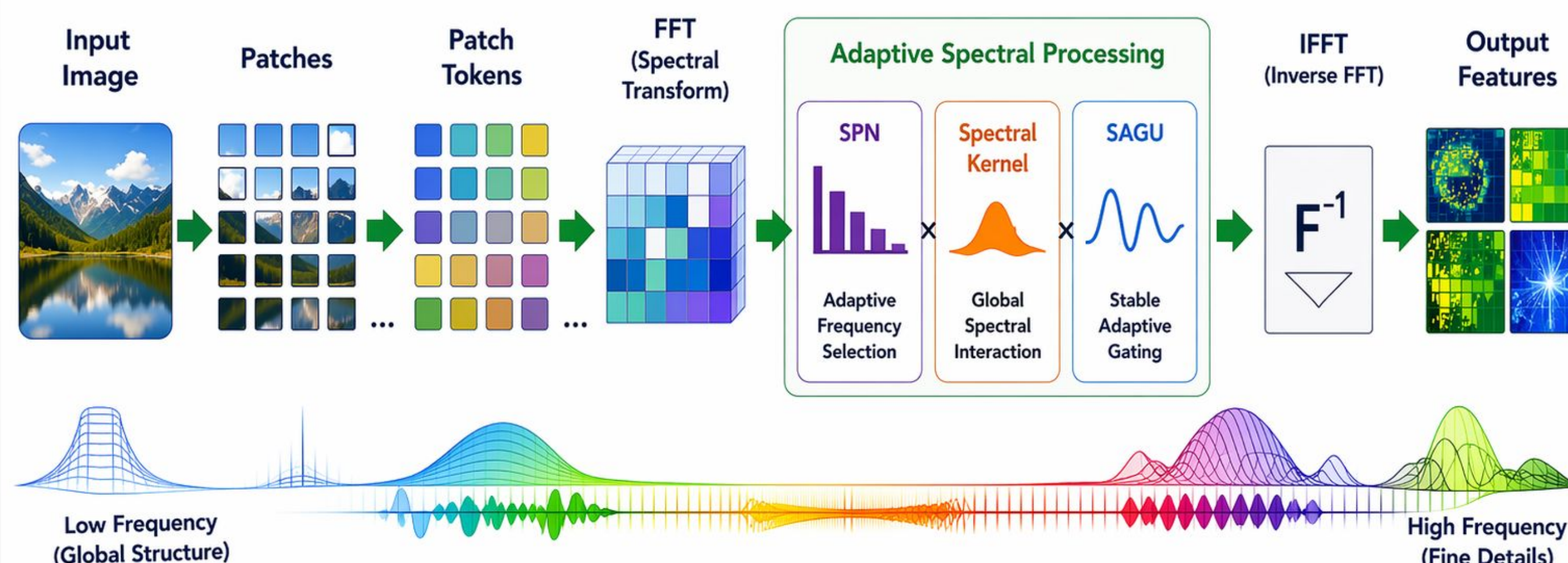
Note:

- Images are not sequences; they are spatial 2D signals.
- Scanning is unnecessary for images because they are not ordered signals.
- HAMSA respects the 2D structure of images by modeling them in the frequency domain, enabling global and parallel understanding without artificial ordering.

HAMSA (Ours)



HAMSA in One View



Images are 2D signals → process globally in the frequency domain, not sequentially.

HAMSA: The Only Scanning-Free, Input-Adaptive Spectral SSM

Unifies SSM Structure + Frequency Domain + Learnable Gating + Input Dependence — with No Scanning

METHOD FAMILY (Examples)	KEY CAPABILITIES					PERFORMANCE @ 224 ² (ImageNet-1K)		
	SSM Structure	Frequency Domain	Learnable Gating	Input-dependent	Scanning Free	Params (M)	FLOPs (G)	Top-1 Acc. (%) ↑
Scan-based SSMs (Vim, VMamba)	✓	✗	✗	✗	✗	~15 – 90	4 – 19	81 – 84.4
Use directional scanning → multiple passes & extra overhead								
Frequency-based Methods (GFNet, FNet)	✗	✓	✗	✗	✓	~15 – 60	4 – 15	81 – 83.7
Operate in frequency domain → but not adaptive								
MambaOut (MambaOut)	✓	✗	✗	✗	✓	~20 – 60	4 – 15	82 – 83.5
Structured & scan-free → but not spectral and not adaptive								
HAMSA (Ours)	✓	✓	✓	✓	✓	72	14.7	84.7 BEST
Unified design: Structured • Spectral • Adaptive • Input-dependent • Scanning-free								



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HAMSA: State-of-the-Art Efficiency Across Model Families

Model Family	Model (Representative)	Type	Params (M)	FLOPs (G)	Latency (ms)	Throughput (img/s)	Memory (GB)	Energy (J)
Convolutional Networks	ResNet-50	CNN	25.5	4.1	9.0	850	3.5	24.5
Transformer-based Models	Swin-B	Transformer	88.0	15.4	15.8	450	8.5	52.5
MLP & Pooling-based Models	PoolFormer	MLP	31.0	5.2	41.2	420	5.5	45.8
State Space Models (SSMs)	VMamba-B	SSM	89.0	18.9	13.2	520	7.2	38.5
HAMSA (Ours)	HAMSA-L	SSM (Scanning-free)	72.0	14.7	9.5	720	4.8	26.8



HAMSA: State-of-the-Art Accuracy with Superior Efficiency

Model	Image Size	Params (M)	MACs (G)	Top-1 Accuracy (%)
Small Scale (≈ 22M params)	TNN [44]	224 ²	6	72.3
	Mamba-2D-S [25]	224 ²	24	81.2
	VMamba-T [30]	224 ²	22	82.2
	LocalVMamba-T [20]	224 ²	26	82.7
	Hamsa-S (Ours)	224 ²	28	83.0
Base Scale (≈ 43-51M params)	VMamba-B [30]	224 ²	44	83.5
	MambaTreeV-S [63]	224 ²	51	84.2
	LocalVMamba-S [20]	224 ²	50	83.7
	Hamsa-B (Ours)	224 ²	43	83.5
Large Scale (≈ 72-241M params)	ViM2-B [1]	224 ²	74	83.9
	VMamba-B [30]	224 ²	89	83.9
	MambaVision-B [15]	224 ²	97	84.2
	Hamsa-L (Ours)	224 ²	72	84.7